PRODUCT REQUIREMENTS DOCUMENT · v1.0

Sasti Dawai

Scan a medicine strip. Find the exact same salt for less.

Problem statement T06 · Aayu 2026 AI Innovation Medical Hackathon
Dr. S.N. Medical College, Jodhpur · Round 1 online, Round 2 on-site (8-hour build)

Stack: React PWA · FastAPI · PostgreSQL (pg_trgm + pgvector) · Vision-LLM + deterministic matcher
Audience: the build team (2–4 people). Read top to bottom once, then use as the build bible.
Date: 13 October 2026

One sentence

Point your phone at any medicine strip and in seconds see the salt, the exact lower-cost equivalents, the monthly saving, and the nearest place to buy it, in Hindi or English, with safety rails that never replace the doctor.

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0 · How to use this PRD

This document is written so a 2–4 person team can start building the same hour they finish reading. It covers the **why** (research), the **what** (features and screens, page by page), the **how** (architecture, ML, data, API), the **safety case**, and the **plan** (8-hour schedule, demo script, risks). Sections 7–11 are the engineering core; sections 14 and 18 are what win the room.

Read this first: the rules. The Aayu rules page says work must be **original and built during the event**, teams are 2–4 people, **no real patient data** may be used, and the build is an **eight-hour on-site** sprint in Jodhpur after an online Round 1 (idea submission). The full rules are only shared with registered teams, so we have not seen the exact wording on pre-written code. **Action:** ask the organisers (aayu@devaarambh.com) what pre-event prep is allowed. Until confirmed, treat this PRD, designs, dataset collection, accounts and API keys as safe prep, and plan to write the application code during the event. The plan in section 15 is built around that assumption.

Document map

Part	Sections	Who reads it
Why and what	1 Executive summary · 2 Research · 3 Goals · 4 Users · 5 Scope · 6 Core logic	Everyone
Experience	7 Screen-by-screen spec · 13 UX and design system	Frontend, design, pitch
Engineering	8 Architecture · 9 ML/AI · 10 Data · 11 API	Backend, ML, data
Trust	12 Safety, ethics, privacy	Everyone (judges will probe this)
Winning	14 Differentiators · 17 Deploy to mobile · 18 Demo and pitch	Everyone
Execution	15 Build plan · 16 QA · 19 Risks · 20 Open decisions · Appendices	Everyone

1 · Executive summary

1.1 The product in one paragraph

Sasti Dawai ("affordable medicine") is a mobile-first web app (installable PWA). A patient or caregiver photographs a medicine strip or box. The app reads the brand, composition, strength, form and printed price; resolves the brand to its **salt(s)**; and finds **exact-equivalent** generics (same salt set, same strength, same dosage form and release type) from the Jan Aushadhi (PMBJP) catalogue. It shows the per-unit price, the **monthly saving** for the patient's actual dose, and the nearest Jan Aushadhi Kendra. Every result ends with the mandatory line: **"Confirm with your doctor or pharmacist."**

1.2 Why this problem is worth solving

- Brand names hide salt names. Patients like Mohanlal (retired teacher, ₹2,800/month on diabetes and BP medicines) cannot read the small print and were never told the salt names.
- India's Jan Aushadhi scheme sells generics at fixed MRPs that are 50–80% below leading brands (some sources say up to 90%), but awareness and discoverability are the bottleneck, not supply.
- An Indian doctor has about two minutes per patient (Aayu's own framing), so the conversation about cheaper equivalents rarely happens. The app fills the gap before and after the consult, without replacing it.

1.3 Why we can win

Judges care about (assumed; confirm on the Submissions page)	Our answer
Real, felt impact	A named persona with a concrete number: ₹2,800 → under ₹600 a month. We show our own real strips saving real rupees on stage.
Working demo on a real phone	Deployed PWA, QR code on the last slide; judges scan with their own phones and try a strip.
Technical depth (backend + ML)	Vision-LLM extraction with barcode/OCR cross-check, brand-to-salt resolver (fuzzy + vector), deterministic exact-match engine, evaluation harness with published accuracy.
Safety and responsibility	Exact-match-only rule, three result tiers, narrow-therapeutic-index warnings, no dosing advice, no real patient data, mandatory disclaimer.
Fit to Aayu's mission	Helps patients understand their own care. Aligns with India's digital health rails (ABHA, ABDM) as a roadmap story.
Execution quality	Clean UX in Hindi and English, offline-tolerant, fast, tested on 50+ strips, clear metrics.

1.4 What we will ship by the end of the event

- A deployed PWA at a public HTTPS URL (installable on Android and iOS home screens).
- Scan → confirm → result flow working end-to-end on real strips, in English and Hindi.
- A catalogue of at least the top ~300 chronic-care brands mapped to salts, matched against the full Jan Aushadhi product list.
- Monthly-basket savings, Kendra finder, shareable summary for the doctor/pharmacist.
- An evaluation report (accuracy on our own strip photo set) shown in the pitch and on an /about page.

2 · Research: the problem, the data, the gap

2.1 The scene (from the problem statement)

Mohanlal, a retired school teacher, spends ₹2,800 a month on diabetes and BP medicines from the chemist near his house. A Jan Aushadhi Kendra two kilometres away sells the exact same salts for under ₹600. He has no idea. The small print on the strips is unreadable, and nobody ever told him the salt names behind the brands.

Implied saving: at least ₹2,200 per month, about **₹26,400 per year** for one patient. That single number anchors the whole pitch.

2.2 Root causes (what the product must attack)

Root cause	Product response
Brand names are marketing; salt names are medicine. Patients know the first, not the second.	Resolve brand → salt automatically and show the salt in large, plain text.
Print on strips is tiny, curved, foil-reflective.	Guided capture with a quality gate, a vision model that handles glare, and a one-tap confirm screen to fix mistakes.
Doctors have about two minutes; nobody explains generics.	Self-serve answer before and after the consult, plus a shareable summary the patient can show the doctor.
Fear that cheaper means lower quality.	Show the verifiable facts: PMBI sources from WHO-GMP certified makers and tests each batch at NABL-accredited labs (as stated by the government). Do not over-claim.
Kendra is far or unknown.	Kendra finder with distance, directions, and an honest note that stock can vary.
Chronic patients buy monthly; savings compound.	Monthly basket view with total saving for the patient's real dose schedule.

2.3 Facts we can cite (with sources)

Fact	Source
Jan Aushadhi basket: 2,110 medicines + 315 surgical items (as reported Aug 2026).	Drishti IAS summary of Ministry data, 1 Aug 2026
17,990 Jan Aushadhi Kendras opened nationwide as of 31 Dec 2025; target 25,000 by March 2027.	Lok Sabha reply reported by Kashmir Life; Government target per Drishti IAS (Dec 2025)
Prices are 50–80% lower than leading brands; fixed MRP nationwide, no additional discounts.	Union Health Minister reply, reported by Swarajya (Mar 2025)
PMBI procures from WHO-GMP certified manufacturers; each batch tested at NABL-accredited labs.	Drishti IAS (Aug 2026), PMBJP scheme documents
Jan Aushadhi Sugam app already exists: locate Kendras, search generics, compare generic vs branded MRP and savings.	BPPI / Department of Pharmaceuticals (via IAS Gyan summary)
An Indian doctor gets about two minutes per patient; 93.95 crore ABHA IDs (Jul 2026); 450 health-tech products on ABDM; SAHI (AI in healthcare strategy) and BODH (health-AI benchmarking platform) launched Feb 2026.	Aayu 2026 About page
Aayu: 8-hour on-site build, teams of 2–4, no real patient data, Round 1 online, Round 2 on-site.	Aayu Problems and Rules pages

Caveat: the Jan Aushadhi product list is published and refreshed by PMBI; counts change. In the app and deck, always show "price as of <date>" and re-pull the list on the morning of the event.

2.4 Existing solutions and the gap

Always confirm with your doctor or pharmacist.

Solution	What it does well	Gap we exploit
Jan Aushadhi Sugam (official app)	Kendra locator, generic search, MRP comparison, savings.	Search-by-typing only: the user must already know the salt or type the brand correctly. No strip scanning, no exact-match safety tiers, no monthly basket from a real prescription, little Hindi/voice help for elderly users (verify by installing it).
Pharmacy e-commerce apps	Large catalogues, "substitute" lists.	Commercially driven; substitutes are usually other brands, not necessarily Jan Aushadhi.
Google Lens / ChatGPT-style photo questions	Can read text.	No verified price database, no exact-match guarantee, hallucination risk on medicines, no Kendra, no audit trail.
Asking the pharmacist	Trusted, in person.	Pharmacist has a commercial incentive to sell the brand; patients often don't ask.

> **Positioning:** we are not "another medicine app". We are the **scan-to-saving layer** on top of the official Jan Aushadhi catalogue, with a strict exact-match safety model. Say this out loud in the pitch: "*Same salt. Same strength. Same form. Lower price. Doctor-confirmed.*"

3 · Goals, non-goals, success metrics

3.1 Product goals

- **G1.** A first-time, low-literacy user gets a correct, safe, understandable result within 10 seconds of tapping Scan.
- **G2.** Zero unsafe matches: never present a non-equivalent as equivalent (false-positive exact matches must be ~0 on our test set).
- **G3.** Make the saving tangible: per-unit price, monthly rupees, yearly rupees, and where to buy.
- **G4.** Work on a mid-range Android phone on 4G in a Tier-2 city, in Hindi and English.
- **G5.** Be demonstrably real: deployed, public URL, real strips, real prices with timestamps.

3.2 Non-goals (say no to these during the 8 hours)

- Diagnosing, recommending doses, recommending switching, or interpreting prescriptions clinically.
- Selling medicines, payments, delivery, or any e-pharmacy flow.
- Reading handwritten prescriptions (roadmap only; high risk, high effort).
- Ayurvedic/homeopathic/OTC cosmetics. Out of scope; the app says so politely.
- User accounts and passwords. Use an anonymous device token; optional phone OTP only as a stretch.

3.3 Success metrics

Metric	Hackathon target	How measured
Field extraction accuracy (brand, salt, strength, form)	≥ 90% per field on 50+ real strip photos	Golden set + eval script (section 9.6)
Exact-match false positives	0 on golden set; any failure is a release blocker	Eval script + manual review by a clinician contact
Scan-to-result latency (p50 / p95)	≤ 6 s / ≤ 10 s on 4G	Server timing + client RUM
Task success in usability test	≥ 4 of 5 non-technical testers finish a scan unaided	Hallway test with 5 people (ideally elderly)
Lighthouse (mobile): Performance / PWA / Accessibility	≥ 85 / pass / ≥ 95	Lighthouse CI
Demo reliability	10/10 successful scans in rehearsal; offline fallback ready	Rehearsal log
Impact framing	₹ saved per user per month shown on result and on a savings screen	App analytics (anonymous)

4 · Users, personas, jobs-to-be-done

Persona	Profile and context	Job to be done	What they fear / need
Mohanlal (primary), 68, retired teacher, diabetes + BP	Pension income, buys monthly from the nearby chemist, WhatsApp-literate, large-print reader, Hindi first.	"Tell me if I'm overpaying and where to buy the same medicine cheaper."	Fears being cheated or getting a weaker medicine. Needs big text, voice, simple words, trust cues.
Sunita (primary), 41, his daughter and caregiver	Manages parents' medicines remotely, uses a smartphone all day.	"Build my parents' monthly basket and see the total saving, then share it with the doctor."	Needs multi-medicine basket, shareable summary, reminders.
Ramesh (secondary), ASHA / community health worker	Visits households in a village, shared phone, patchy network.	"Check a family's strips quickly during a home visit."	Needs offline tolerance, fast flow, Hindi, no login.
Kendra owner / pharmacist (secondary)	Runs a Jan Aushadhi Kendra; wants footfall.	"Show patients my shop is the cheap equivalent."	Needs accurate stock expectations. Future B2B hook.
Judge / doctor (stakeholder)	Clinician or technologist evaluating in minutes.	"Is this safe, real, and better than what exists?"	Will test edge cases and ask about errors. Needs visible safeguards.

4.1 Primary user journey (Mohanlal)

- Opens the app link from a WhatsApp message his daughter sent, or the QR code. Taps **Add to Home Screen**.
- Chooses Hindi. Taps the big **Scan strip** button, points the camera, a frame guides him.
- Within seconds sees: "This is **Metformin 500 mg** (tablet). You pay ₹X per month. Exact same medicine at Jan Aushadhi costs ₹Y."
- Taps **Add to my monthly list**, repeats for 4 more strips. Savings screen shows ₹ per month and per year.
- Taps **Share with doctor** → one-page summary on WhatsApp/PDF. Taps **Nearest Kendra** → map and directions.
- Every screen ends with: **Confirm with your doctor or pharmacist.**

5 · Scope and functional requirements

Priorities use MoSCoW: **M** must (demo fails without it), **S** should (strong boost), **C** could (only if time). IDs are used in the task board.

ID	Feature	Pri	Acceptance criteria (testable)
F-01	Camera scan of strip or box (front or back), guided capture, multi-photo (up to 3)	M	On a mid-range Android phone over HTTPS, the camera opens in ≤ 2 s; user can retake; uploads compress to ≤ 1.5 MB per image.
F-02	Image quality gate (blur, glare, too dark, too far)	M	Blurry or dark photo triggers a plain-language prompt ("Hold steady, move closer") before upload.
F-03	Extraction of brand, salts, strength, form, release type, pack size, MRP, expiry	M	Returns strict JSON matching the schema; nulls for unreadable fields; never invents values.
F-04	Barcode/QR read where present	S	If a GS1/QR code decodes, use it to fetch the exact product; fall back to vision if not found. (Verify coverage on real packs.)
F-05	Confirm/edit screen with confidence flags	M	Low-confidence or cross-check-failed fields are highlighted; user can edit; nothing proceeds to matching without confirmation when confidence is low.
F-06	Brand → salt resolver (DB + fuzzy + vector)	M	Top-1 resolution ≥ 95% on curated brand list; returns canonical salt IDs; unresolved brand shows "We couldn't verify this" and no match.
F-07	Exact-equivalent matcher with tiers (Exact / Close / None)	M	Exact tier requires identical salt set, strength, dosage form and release type. Close tier is never counted in savings and carries a stronger warning.
F-08	Savings calculator (per unit, per month, per year)	M	Uses normalised per-unit price × units/month; user can set dose frequency; shows "price as of <date>".
F-09	Result screen with plain-language explanation	M	Salt in large text, brand vs generic prices, saving in rupees, tier badge, disclaimer pinned at bottom.
F-10	Kendra finder (map, distance, directions, hours if known)	M	Shows ≥ 3 nearest Kendras for the user's location or typed PIN; works without GPS via PIN/city.
F-11	My Monthly List (basket) with total saving	M	Add/remove medicines, edit units/month; total monthly and yearly saving updates instantly; persisted on device.
F-12	Hindi + English UI, voice read-aloud of result	S	Language toggle on every screen; Web Speech API reads result in hi-IN or en-IN; text scale control.
F-13	Share with doctor/pharmacist (PDF and WhatsApp text)	S	One-page PDF listing scanned meds, salts, strengths, generic options, disclaimer; opens WhatsApp share sheet.
F-14	Duplicate-salt alert in basket	S	If two items share a salt, show "Same medicine under two brands. Ask your doctor." (clinically valuable, low effort).
F-15	Narrow-therapeutic-index (NTI) and special-form warnings	M	For a curated list (e.g. levothyroxine, warfarin, antiepileptics), show "Do not switch without your doctor" and suppress savings emphasis.
F-16	Offline tolerance and PWA install	S	Installable; app shell + basket + last Kendras cached; scan queued with a clear message when offline.
F-17	Data freshness and provenance page	S	Shows data sources, last-updated dates, counts, and evaluation results.
F-18	Feedback "Is this wrong?" button on every result	S	Sends scan ID + correction; stored without personal data.
F-19	WhatsApp bot entry via n8n	C	User sends a strip photo to a number/sandbox; receives result text. Only if core is stable.
F-20	ABHA/FHIR export (mock, sandbox)	C	Exports basket as a FHIR MedicationStatement bundle for roadmap storytelling.

Cut line: If at hour 5 F-01…F-11 and F-15 are not stable on a real phone, stop adding features and harden. A smaller, flawless demo beats a larger flaky one.

6 · Core logic: what "exact equivalent" means

This section is the product's brain. It must be implemented as **deterministic code** (not an LLM guess) and reviewed by someone with clinical judgment.

6.1 Canonical medicine model

Every product (brand or generic) is normalised to the tuple below. Two products are comparable only if every field matches.

Field	Meaning	Example
salt set	Set of active moieties (order-independent) with their strengths	{Metformin 500 mg}; {Glimepiride 2 mg + Metformin 500 mg}
strength	Value + unit normalised (mg, mcg, g, IU, % w/v, mg/5 ml)	500 mg; 125 mg/5 ml
dosage form	Tablet, capsule, syrup, suspension, injection, gel, drops...	tablet
release type	IR, SR, ER, DR/EC, MR, ODT, chewable, dispersible	SR (sustained release) is **not** equal to IR
route	Oral, topical, parenteral, ophthalmic...	oral
pack and unit	Count per strip/bottle; price per unit = MRP ÷ count	10 tablets, ₹30 → ₹3.00 per tablet

6.2 Result tiers

Tier	Rule	UI treatment	Counts toward savings?
EXACT	Same salt set, same strength per unit, same dosage form, same release type, same route.	Green badge "Exact same medicine"	Yes
CLOSE	Same salt set but different strength, or same strength with different release type/pack logic (e.g., 2 × 250 mg for 500 mg).	Amber badge "Similar. Ask your doctor before switching"	No (shown separately)
NONE	No Jan Aushadhi product with the same salt set, or brand could not be verified.	Neutral card "No cheaper exact match found" (never hide this; honesty builds trust)	No

6.3 Savings formula

```
unit_price_brand   = brand_mrp / brand_pack_count          # e.g. 120 / 10 = 12.00 per tablet
unit_price_generic = generic_mrp / generic_pack_count      #       30 / 10 =  3.00 per tablet
units_per_month    = doses_per_day * 30                    # user-set; default from label only if the user confirms
monthly_saving     = (unit_price_brand - unit_price_generic) * units_per_month
yearly_saving      = monthly_saving * 12                   # illustrative arithmetic, not real prices
# Example (ILLUSTRATIVE): (12.00 - 3.00) * 60 = ₹540 per month for a twice-daily tablet.
# Syrups/gels: normalise to per ml / per g. Injections: per vial/ampoule. Never mix units.
```

Rule: savings are displayed only for EXACT matches, with the price date. The app never says "stop taking X". It says "the same salt is available at Y; confirm with your doctor or pharmacist."

6.4 Edge cases and required behaviour

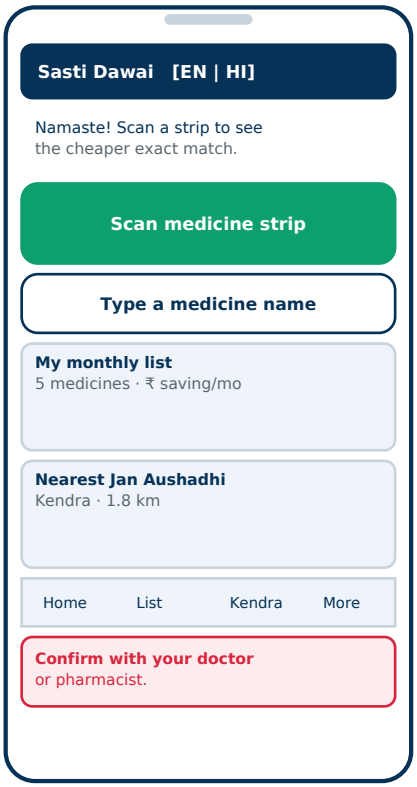
Case	Required behaviour
Sustained/extended-release (SR/ER/XR/MR) vs immediate release	Different release types are never EXACT. Show CLOSE with warning if the generic exists.
Fixed-dose combinations (e.g., two or three salts)	EXACT only if all salts and strengths match. Partial overlap is CLOSE at best.
Different strength (e.g., 40 mg vs 20 mg)	CLOSE. Never auto-suggest splitting or doubling tablets.

Case	Required behaviour
Different salt form (e.g., hydrochloride vs besylate of the same moiety)	Mark "Pharmacist to confirm" and downgrade to CLOSE unless a clinician-approved equivalence table says otherwise.
NTI / critical drugs (e.g., levothyroxine, warfarin, phenytoin, carbamazepine, digoxin, lithium, cyclosporine, tacrolimus; list to be clinically reviewed)	Show "Do not switch without your doctor" prominently; savings shown in grey, no green "Exact" celebration.
Prescription-only (Schedule H/H1/X) and controlled drugs	Allow lookup but add "Prescription medicine" note; never imply the app authorises purchase.
Antibiotics and short courses	Show match but remind to complete the prescribed course; do not compute monthly saving.
No Jan Aushadhi equivalent found	Say so plainly. Optionally show other approved lower-priced brands only if clearly labelled (stretch).
Unreadable or partial strip, foreign pack, non-allopathic product	Ask for another photo or manual search; never guess.
Expired strip (expiry read)	Warn "This strip looks expired. Do not use. Ask your pharmacist."
Brand and printed composition disagree (OCR/VLM vs DB)	Block automatic matching; ask the user to confirm which is correct.

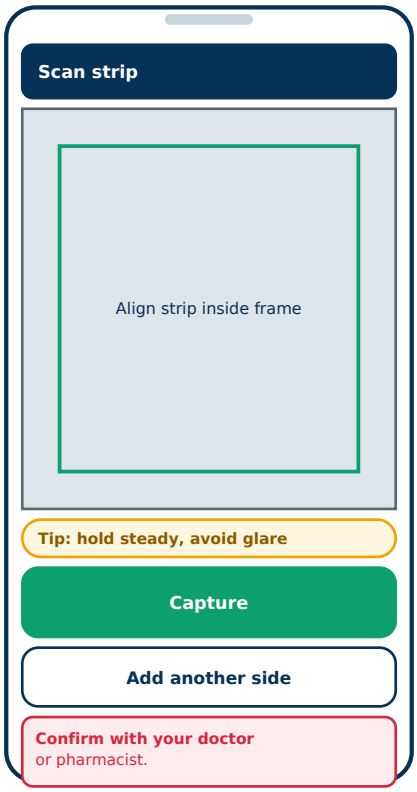
7 · Screen-by-screen specification

Mobile-first (360×800 baseline). Every screen shows the persistent safety line at the bottom. The six wireframes below are the critical path; the table after them specifies all 12 screens.

S02 Home



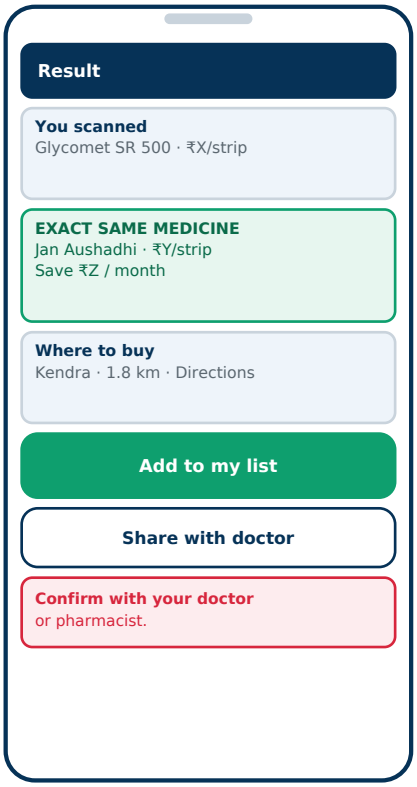
S03 Scan



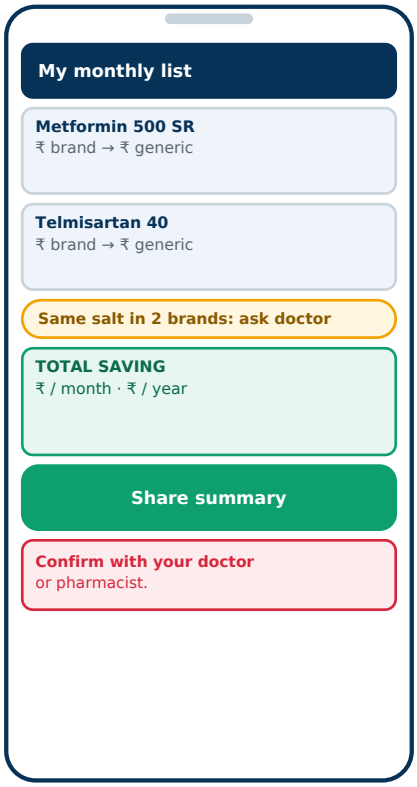
S04 Confirm



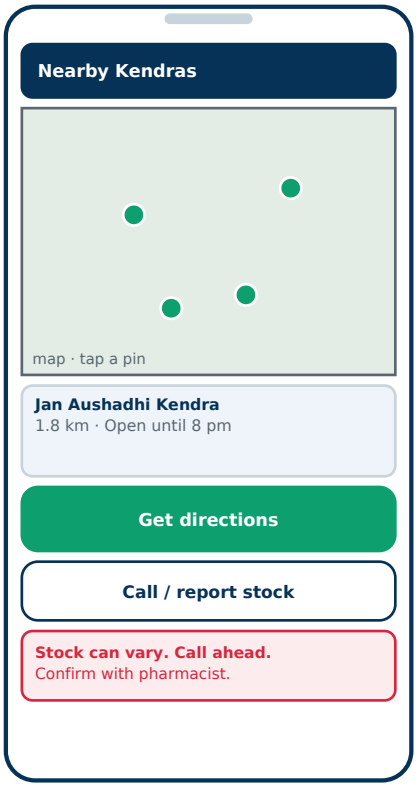
S05 Result



S07 My monthly list



S08 Kendras



S01 · Splash and language

Purpose	Set language and build trust in 5 seconds.
Content	App name, language chooser (English / Hindi), one-line promise, short consent note ("Photos are used only to read the strip; not stored by default").
Actions	Choose language → Continue. Remember choice in localStorage.
States	Loading logo only; no network needed.
Acceptance	First launch only; language persists; ≤ 1 tap to continue.

S02 · Home

Purpose	One obvious action: scan.
Content	Big Scan button, secondary "Type a medicine name", My monthly list card with running saving, nearest Kendra card, bottom tab bar.
Actions	Scan, search, open list, open Kendras, change language/text size.
States	Empty list shows a friendly example; offline banner when offline.
Acceptance	Scan button reachable with one thumb; text ≥ 18 px by default.

S03 · Scan (camera)

Purpose	Capture a readable photo.
Content	Live camera with guide frame, flash toggle, tip chip, Capture, "Add another side" (up to 3), gallery upload fallback.
Actions	Capture, retake, add side, upload from gallery.
States	Permission denied → explain how to enable + gallery fallback. Quality gate failure → inline tip. Offline → queue.
Acceptance	Blur/glare detection triggers tip; camera permission handled gracefully on iOS Safari and Chrome Android.

S04 · Confirm and edit

Purpose	Human-in-the-loop safety check.
Content	Extracted brand, salts, strength, form, release, pack size, MRP; confidence flags; doses-per-day stepper; "Is this correct?".
Actions	Edit any field, retake photo, confirm.
States	Low-confidence fields highlighted amber; cross-check mismatch blocks Confirm until resolved.
Acceptance	No result is computed from low-confidence data without confirmation.

S05 · Result (the money screen)

Purpose	Deliver the saving and the safe next step.
Content	Scanned medicine card; tier badge (Exact/Close/None); generic option(s) with unit price and strip price; monthly + yearly saving; where to buy; plain explanation; "price as of" date; pinned disclaimer; "Is this wrong?" link.
Actions	Add to list, share, directions, read aloud, report wrong, view all options.
States	NONE tier shows honest empty state; NTI drugs show "Do not switch without your doctor"; error state offers manual search.
Acceptance	Saving math reproducible by hand; disclaimer cannot be scrolled out of view.

S06 · Alternatives detail

Purpose	Transparency for sceptical users and judges.
Content	Side-by-side comparison table (salt, strength, form, release, pack, unit price), source and date, tier explanation.
Actions	Back, add specific option to list.
States	None.
Acceptance	Every cell traceable to a DB row.

S07 · My monthly list

Purpose	Make the saving accumulate.
Content	List of medicines with brand price vs generic price, units/month steppers, duplicate-salt alert, total monthly and yearly saving, share button.
Actions	Edit units, remove, share, clear list.
States	Empty list explains how to add; stored locally; offline OK.
Acceptance	Totals update instantly; duplicate-salt alert appears when two items share a salt.

S08 · Kendra finder

Purpose	Turn savings into action.
Content	Map with pins, list sorted by distance, address, hours if known, directions deep link, call button, "Stock can vary" note.
Actions	Use my location, enter PIN/city, directions, call.
States	Location denied → PIN entry. No Kendra within radius → widen radius.
Acceptance	≥ 3 results for any Jodhpur PIN; directions open in Google Maps.

S09 · Share with doctor / pharmacist

Purpose	Bridge to the human who confirms.
Content	One-page summary: patient-entered medicines, salts, strengths, generic options, prices with date, disclaimer, QR back to the result page.
Actions	Download PDF, share to WhatsApp, copy text.
States	Generation failure → plain-text fallback.
Acceptance	PDF fits one page for up to 8 medicines; no personal identifiers unless user types them.

S10 · Search by name

Purpose	Fallback when scanning fails.
Content	Search box with typeahead (brand or salt), recent searches, results with tier badges.
Actions	Select a result → goes to Confirm/Result.
States	No match → suggest scan or ask pharmacist.
Acceptance	Fuzzy search handles common misspellings and Roman-Hindi brand names.

S11 · About, safety and data

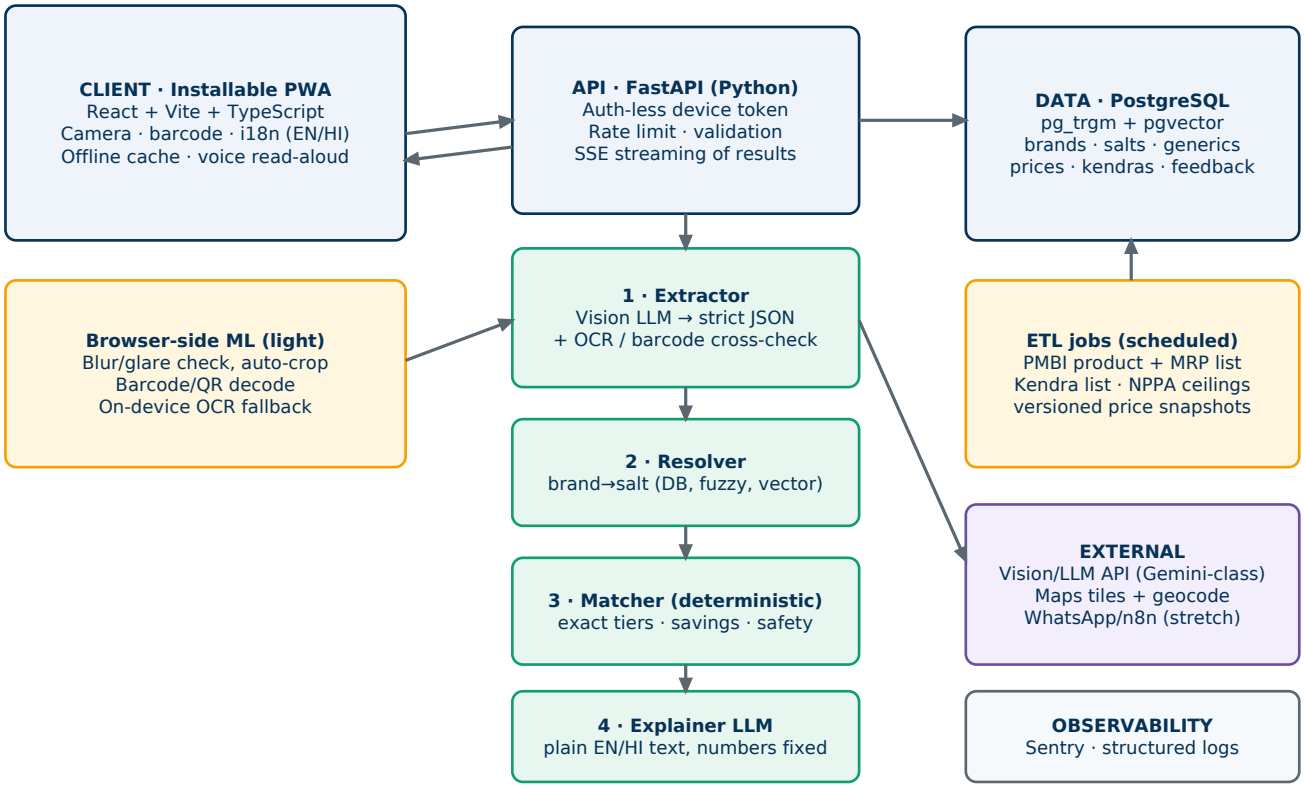
Purpose	Credibility page for judges.
Content	How it works, data sources and last-updated dates, evaluation results, safety rules, privacy note, contact, feedback link.
Actions	Open sources, send feedback, change settings.
States	None.
Acceptance	Every claim on the page links to a source or an eval report.

S12 · Settings

Purpose	Accessibility and control.
Content	Language, text size (Normal/Large/Extra large), read-aloud default, clear local data.
Actions	Toggle, clear data.
States	None.
Acceptance	Text size applies app-wide immediately.

8 · Architecture and technology stack

Principle: **use ML where perception is needed, and deterministic code where safety is needed.** The vision model reads the strip; a database and a rule engine decide equivalence and compute money. An LLM is never the source of a price or an equivalence claim.



8.1 Recommended stack

Layer	Choice	Why	Alternative
Frontend	React + Vite + TypeScript, Tailwind CSS, shadcn/ui, TanStack Query, react-i18next, vite-plugin-pwa (Workbox)	Team already knows React/Vite (existing project). Fast dev, small bundle, PWA install.	Next.js if SSR/SEO is needed (not needed here)
Camera and on-device	MediaDevices getUserMedia, canvas for crop/compress, a barcode decoder library, Laplacian-variance blur check	Runs in browser, no native app needed.	Capacitor wrapper for an APK as backup
Backend API	Python FastAPI + Pydantic v2 + Uvicorn, async SQLAlchemy	Python keeps ML, data and API in one language; strict schemas; auto OpenAPI docs.	Node/Fastify if no ML in backend
Database	PostgreSQL with pg_trgm (fuzzy) and pgvector (embeddings)	One DB for relational data, fuzzy and vector search. Managed free tiers exist (Neon, Supabase).	SQLite + FTS for the simplest demo
Vision extraction	Multimodal LLM (Gemini Flash-class; Google for Developers is an Aayu community partner) with JSON-schema output	Handles curved foil text and glare better than classic OCR; structured output reduces parsing bugs.	Claude vision-capable model; PaddleOCR/Tesseract.js fallback
Embeddings / fuzzy	rapidfuzz + trigram first; small multilingual embedding model (e.g., e5-small class) for vector fallback	Brand names are noisy; combining lexical and semantic retrieval improves recall.	Trigram only (fine if time is short)

Layer	Choice	Why	Alternative
Maps	MapLibre GL with OpenStreetMap tiles (or Google Maps JS if key available); geocoding by PIN	No-cost and lightweight; directions deep-link to Google Maps.	Leaflet
Workflow / bot	n8n (Aayu AI partner) for WhatsApp-style flows (stretch)	Aligns with sponsors; quick no-code integration.	Twilio/WhatsApp Cloud API direct
Hosting	Frontend: Vercel or Cloudflare Pages. Backend: Google Cloud Run (or Render/Fly). DB: Neon/Supabase.	HTTPS by default (camera needs it), free tiers, instant deploys.	Single VPS with Docker Compose
Observability	Sentry (frontend + backend), structured JSON logs, a /health endpoint	Catch demo-day failures fast.	Console logs only
CI/CD	GitHub Actions: lint, tests, deploy on main	One-click rollback story.	Manual deploy (risky)

8.2 Repository layout

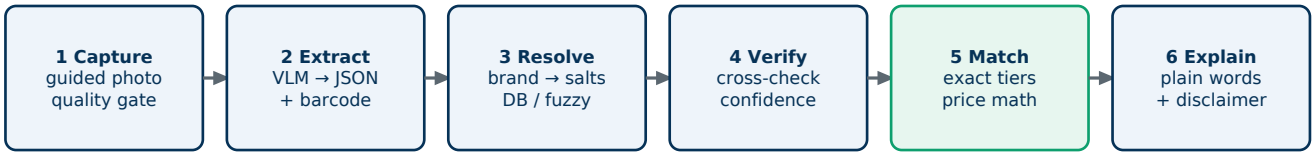
```
sasti-dawai/
├─ apps/
│  ├─ web/                # React PWA (Vite)
│  │   ├─ src/screens/     # S01..S12
│  │   ├─ src/lib/         # api client, i18n, camera, quality-gate
│  │   └─ public/          # manifest, icons, offline.html
│  └─ api/                # FastAPI
│      ├─ app/routers/     # scan, match, medicines, kendras, report
│      ├─ app/services/    # extractor, resolver, matcher, explainer
│      ├─ app/models/      # SQLAlchemy + Pydantic schemas
│      └─ app/prompts/     # versioned prompts + JSON schema
├─ data/
│  ├─ raw/                 # PMBI list, Kendra list, NPPA, curated brands (CSV)
│  ├─ etl/                 # load + normalise scripts
│  └─ eval/                # golden strip photos + labels + eval.py
├─ docs/                   # this PRD, API docs, eval report
└─ .github/workflows/      # ci.yml, deploy.yml
```

8.3 Non-functional requirements

Area	Requirement
Performance	p50 scan→result ≤ 6 s on 4G. Compress images client-side to about 1280 px long edge. Stream partial results (SSE) so the UI shows progress.
Reliability	Retry once on vision API failure; fall back to OCR + barcode; final fallback is manual search. Keep backend warm (cron ping) to avoid cold starts.
Security	HTTPS only, strict CORS, rate limiting per device token and IP, input validation, secrets in env vars, no keys in the frontend.
Privacy	Images deleted after processing by default (or TTL ≤ 1 hour if kept for debugging with consent); no PII; anonymous device token; analytics without identifiers.
Accessibility	WCAG 2.2 AA contrast, 18 px base text, 48 px tap targets, screen-reader labels, text-scale control, voice read-aloud.
Compatibility	Chrome Android 10+ (primary), Safari iOS 16+ (secondary); test on at least 3 real devices.
Scalability story	Stateless API behind autoscaling; read-heavy DB with indexes; nightly ETL; cache hot lookups.

9 · Machine learning and AI design

9.1 End-to-end pipeline



Stage	Technique	Output	Failure handling
1 Capture	Client-side quality gate: Laplacian variance (blur), brightness histogram (dark), saturation-clipping (glare)	Up to 3 compressed images	Prompt user to retake
2 Extract	Vision LLM with strict JSON schema + temperature 0; in parallel decode barcode/QR if present	Brand, printed composition, strength, form, release, pack, MRP, expiry, per-field confidence	Retry; OCR fallback; manual search
3 Resolve	Exact name lookup → trigram fuzzy (rapidfuzz/pg_trgm) → vector similarity; salts normalised to canonical IDs	Canonical product + salt set from DB (source of truth)	Unresolved → ask user
4 Verify	Compare DB composition vs printed composition read by the model; compute combined confidence	Verified / needs-confirmation flag	Mismatch blocks matching
5 Match	Deterministic rule engine (section 6)	Tiered alternatives + prices + savings	NONE tier if no candidate
6 Explain	LLM writes 2–3 plain sentences in EN/HI from structured facts only; numbers injected by code	Human-friendly explanation + fixed disclaimer	Template fallback (no LLM)

9.2 Where ML is real here (and why that is a feature)

- **Perception (vision model):** reading curved, reflective, multilingual strip text is a genuine ML problem; we use a multimodal model with schema-constrained output.
- **Entity resolution (fuzzy + embedding retrieval):** brand names have spelling variance and OCR noise; we blend lexical and vector search and measure top-1/top-3 accuracy.
- **Calibrated confidence:** combine model self-reported field confidence, cross-check agreement, and retrieval score into one gate; tune the threshold on the golden set to hit zero unsafe passes.
- **Evaluation harness (differentiator):** a reproducible script reporting field accuracy, resolution accuracy, false-positive rate and latency, in the spirit of India's BODH benchmarking idea.
- **Deterministic safety layer:** equivalence and money are code, not generation. We say this explicitly to judges as a design choice for medical safety.

9.3 Stretch ML (only if core is done)

- Fine-tune or train a small strip-text-region detector (YOLO-class) to crop composition lines before extraction; compare accuracy/latency vs full-image VLM.
- On-device model (WebGPU/ONNX Runtime Web) for blur/glare classification and brand-name OCR to enable true offline scanning.
- Active learning loop: user corrections on "Is this wrong?" become labelled examples for the next eval run.

9.4 Extraction prompt and schema (v1)

```
SYSTEM: You read photos of Indian medicine strips and boxes. Transcribe ONLY what is printed and visible.
Rules: (1) Never infer or guess a salt from the brand name. (2) If a value is unreadable, return null.
(3) Return JSON only, matching the schema. (4) Give a 0-1 confidence per field. (5) If the image is not a
medicine pack, set is_medicine_pack=false.

SCHEMA:
{
 "is_medicine_pack": bool,
 "brand_name": string|null,
 "manufacturer": string|null,
 "composition": [{"salt": string, "strength_value": number, "strength_unit": "mg|mcg|g|IU|ml|%",
                  "per": "tablet|capsule|5ml|ml|g|dose"}],
 "dosage_form": "tablet|capsule|syrup|suspension|injection|gel|cream|drops|inhaler|other",
 "release_type": "IR|SR|ER|DR|MR|ODT|chewable|dispersible|unknown",
 "pack_size": {"count": number, "unit": "tablets|capsules|ml|g|vials"},
 "mrp_inr": number|null, "expiry": "YYYY-MM"|null, "rx_schedule": "H|H1|X|OTC|unknown",
 "legibility": "good|partial|poor",
 "field_confidence": {"brand_name": 0-1, "composition": 0-1, "dosage_form": 0-1, "mrp_inr": 0-1}
}
```

9.5 Explainer prompt guardrails

- Input is a JSON of verified facts. The model may only rephrase; it may not add medical claims, doses, or advice to switch.
- Output must include the fixed closing sentence (added by code, not by the model).
- Reject any output containing words such as "stop taking", "safe to replace", "better than" (post-filter).

9.6 Evaluation plan

Item	Plan
Golden set	Team members photograph 50–100 strips/boxes from home (no patient data). Include foil-only strips, boxes, curved bottles, glare, low light, Hindi text. Label ground truth by hand; have a clinician contact verify the 20 most important.
Metrics	Per-field accuracy; brand→salt top-1/top-3; exact-match precision (must be 100% on set); recall; p50/p95 latency; cost per scan.
Error taxonomy	Misread strength, wrong release type, wrong brand, missing combination salt, glare failure. Track counts; fix top 2 before demo.
Report	Auto-generate a Markdown/HTML eval report each run; surface summary on S11 and one slide in the deck.

10 · Data: sources, schema, pipeline

10.1 Data sources

Dataset	Use	Source and notes	Risk
Jan Aushadhi (PMBJP) product list: drug code, generic name, unit size, MRP, group	Generic catalogue and prices (the backbone)	Official: janaushadhi.gov.in / PMBI product list (refreshed regularly; some third-party mirrors state daily updates). Re-download the morning of the event.	Format changes; parse defensively
All-India Kendra list (name, address, PIN, state, district)	Kendra finder	Official downloadable Kendra list on the Jan Aushadhi site (verify current link). Geocode Jodhpur-area addresses ahead of time.	Geocoding gaps
Brand → salt mapping (top ~300 chronic and common brands)	Resolver	Curate by hand from product labels and public references; verify with a clinician contact. Public scraped datasets exist but check licence and accuracy before use; avoid scraping commercial pharmacy sites (terms of service).	Accuracy and licensing
NPPA ceiling prices (scheduled formulations)	Optional sanity check on brand MRPs	NPPA/DPCO notifications (public).	Parsing effort
Printed MRP on strip	Brand price (per scan)	Read from the pack by the extractor; user can edit.	Misread price
Our own golden set photos	Evaluation	Created by the team.	None

Reality check (do this the day before): visit a Jan Aushadhi Kendra in Jodhpur, photograph 10 shelf prices, and buy or photograph 5 branded strips for the same salts. Real receipts and real strips make the demo credible and let us validate our price data against the shop.

10.2 Relational schema (PostgreSQL)

```
salts(id, name, moiety, aliases[], is_nti bool, notes)
products(id, kind ENUM('brand','generic'), name, manufacturer, dosage_form, release_type, route,
         pack_count, pack_unit, mrp_inr, rx_schedule, source, source_row, price_date, embedding vector(384))
product_salts(product_id, salt_id, strength_value, strength_unit, per_unit)       -- the composition set
equivalence(brand_product_id, generic_product_id, tier ENUM('exact','close'), reason, reviewed_by, reviewed_at)
kendras(id, name, address, pin, district, state, lat, lon, phone, hours, source, geocoded_at)
price_snapshots(product_id, mrp_inr, snapshot_date)                                 -- history and provenance
scans(id, device_token, created_at, status, tier, latency_ms, extractor_version)   -- no images, no PII
feedback(id, scan_id, field, wrong_value, right_value, created_at)
data_sources(id, name, url, fetched_at, row_count, sha256)
-- indexes: GIN trigram on products.name and salts.aliases; ivfflat/hnsw on products.embedding;
--          btree on (dosage_form, release_type); GiST/earthdistance (or PostGIS) on kendras(lat,lon)
```

10.3 ETL steps

- **Ingest:** download PMBI list → store raw file + checksum in data_sources.
- **Parse and normalise:** split salt strings ("+", "and"), normalise units (mg/mcg), detect form (tablet, capsule, syrup...), detect release (SR/ER/XR/MR/DR/EC), compute pack size and unit price.
- **Map salts:** canonicalise names (strip "hydrochloride", "sodium" into salt-form attribute but keep it); build the aliases list.
- **Build equivalence:** for each curated brand, compute exact/close candidates with the section 6 rules; store reason strings. Human-review the top 100 pairs.
- **Geocode Kendras:** prioritise Rajasthan/Jodhpur; fall back to PIN centroid.
- **Embed:** create embeddings for product names (and salt aliases) for vector fallback.
- **Validate:** unit tests on parser; spot-check 30 rows against the source PDF/website; fail the build if row counts drop unexpectedly.

10.4 Data governance

- Show source name and "price as of" date wherever a price appears.
- Keep raw files and checksums so any number can be traced.
- Do not store scan images by default; keep scans table anonymous.
- No real patient data in development or demo (Aayu rule). Use team members' own, non-sensitive medicine strips.

11 · API specification (v1)

Method and path	Purpose	Request	Response (key fields)
POST /v1/scan	Extract and resolve from photos	multipart: images[1..3], lang	scan_id, extracted{...}, resolved{product_id, salts[]}, confidence, needs_confirmation, flags[]
POST /v1/match	Find equivalents and savings	JSON: product_id or composition, units_per_month	tier, alternatives[{product, unit_price, strip_price, saving_month, saving_year, source, price_date}], warnings[]
GET /v1/medicines/search	Typeahead by brand or salt	q, limit	results[{id, name, salts, form, tier_hint}]
GET /v1/medicines/{id}	Product detail	id	product, composition, form, release, price, source
GET /v1/kendras	Nearby Kendras	lat,lng or pin; radius_km	kendras[{name, address, distance_km, lat, lon, phone, hours}]
POST /v1/basket/summary	Basket totals	items[{product_id, units_per_month}]	per-item savings, total_month, total_year, duplicate_salts[]
POST /v1/report	Generate share summary	items[], lang	pdf_url or base64, whatsapp_text
POST /v1/explain	Plain-language text	facts JSON, lang	text (LLM), disclaimer (fixed)
POST /v1/feedback	Flag wrong result	scan_id, field, correction	ok
GET /v1/meta/data-health	Provenance and eval	—	sources[{name, fetched_at, rows}], eval{field_acc, fp_rate, n}
GET /health	Liveness	—	status, version

11.1 Sample: POST /v1/match response

```
{
  "tier": "exact",
  "scanned": {"name": "Glycomet SR 500", "salts": [{"salt": "Metformin", "value": 500, "unit": "mg"}],
              "form": "tablet", "release": "SR", "strip_price": 40.0, "pack_count": 10},
  "alternatives": [{
     "product_id": "gen_000123", "name": "Metformin Prolonged Release Tablets IP 500 mg (Jan Aushadhi)",
     "strip_price": 12.5, "pack_count": 10, "unit_price": 1.25,
     "saving_month": 165.0, "saving_year": 1980.0, "units_per_month": 60,
     "source": "PMBI product list", "price_date": "2026-10-12"}],
  "warnings": [],
  "disclaimer": "Confirm with your doctor or pharmacist."
}
// NOTE: numbers above are illustrative placeholders to show the shape, not real prices.
```

11.2 Conventions

- Pydantic models are the contract; generate OpenAPI and a typed TypeScript client for the frontend.
- Errors use RFC 7807-style problem JSON with a stable *code* (e.g., LOW_CONFIDENCE, UNRESOLVED_BRAND, VISION_UNAVAILABLE).
- All responses include *disclaimer* and *data_version*.
- Idempotent scan retries with a client-generated request ID.

12 · Safety, ethics, privacy and compliance

Prime directive: the app helps people *ask better questions* of their doctor or pharmacist. It never tells anyone to stop, switch, or change a dose.

12.1 Safety rules (non-negotiable)

- Disclaimer "Confirm with your doctor or pharmacist" appears on every result screen, every shared PDF/WhatsApp message and the footer of every page (required by the problem statement).
- Only EXACT matches show savings; CLOSE matches are labelled and excluded from totals.
- No dosing, no "equivalent dose" suggestions, no tablet splitting, no interaction diagnosis.
- NTI and critical-drug list shows a "Do not switch without your doctor" banner (list reviewed by a clinician).
- Low-confidence reads require user confirmation; the app prefers saying "I'm not sure" to guessing.
- Wrong-result reporting is one tap; reports are reviewed before demo.

12.2 Honesty about generics

- State what is verifiable: the government says PMBI sources from WHO-GMP certified manufacturers and tests each batch at NABL-accredited labs. Do not claim bioequivalence guarantees beyond sources.
- Say that availability can vary by Kendra and that prices can change; always show the price date.
- Do not disparage brands or pharmacists. Frame it as informed choice.

12.3 Privacy and data protection

- Data minimisation: no names, phone numbers or health records are required. Anonymous device token only.
- Photos are processed and discarded by default; if kept for debugging, only with consent and a short TTL.
- Aligns with India's Digital Personal Data Protection Act, 2023 principles (notice, purpose limitation, minimisation). Add a short plain-language privacy notice on S01 and S11. This is not legal advice; have a mentor review.
- Third-party processors (vision/LLM API, hosting, maps) are listed on S11. Avoid sending identifiers to them.
- Hackathon rule: **no real patient data**. Use only the team's own non-sensitive strips and mock baskets.

12.4 Responsible-AI checklist for the pitch

Question a judge may ask	Our answer
Can the AI hallucinate a medicine?	Extraction is schema-constrained, cross-checked against the database and gated by confidence; matching and prices are deterministic.
What if it's wrong?	Confirm screen, tiers, disclaimer, one-tap reporting, and a measured false-positive rate on our test set.
Does it replace the doctor?	No. It prepares the patient for the doctor and pharmacist conversation and ends every result with that instruction.
Whose data is this?	Public government price list and our own curated mapping; no patient data stored.

13 · UX and design system

13.1 Design principles

- **Elder-first:** base text 18 px, 48 px tap targets, high contrast, one primary action per screen.
- **Plain words:** say "medicine" and "salt (the active ingredient)" with an info tooltip; avoid jargon.
- **Show the money:** rupees in large type, green for savings; but calm for NTI/Close tiers (no celebration).
- **Trust cues:** source and date next to every price; visible disclaimer; honest empty states.
- **Voice and language:** Hindi and English everywhere; read-aloud on the result screen.
- **Fast feedback:** skeleton loaders and streamed steps ("Reading strip… Finding salt… Checking Jan Aushadhi…").

13.2 Visual language

Token	Value	Usage
Primary navy	#063257 (matches Aayu's theme colour)	Headers, text, brand
Savings green	#0E9F6E	Exact tier, saving amounts, primary buttons
Caution amber	#F4A100	Close tier, low confidence
Safety red	#D7263D	Disclaimer strip, NTI warnings, errors
Surface	#EEF4FA / white	Cards and backgrounds
Type	A highly legible sans (e.g., Inter) with a Devanagari companion (e.g., Noto Sans Devanagari); numerals large and tabular	All text
Shape and motion	12 px radius, subtle elevation, reduced-motion respected	Components

Your portfolio uses a neo-brutalist look; for this product, keep a calm, trustworthy medical feel instead, with the neo-brutalist boldness reserved for big numbers and the hero button.

13.3 Microcopy (English with Roman-Hindi reference; ship real Devanagari in i18n files)

Moment	English	Hindi (Roman script)
Hero	Scan your medicine strip. See the same medicine for less.	Apni dawai ki patti scan karein. Wahi dawai kam daam mein dekhein.
Exact match	Exact same medicine: same salt, same strength, same form.	Bilkul wahi dawai: wahi salt, wahi taakat, wahi roop.
Saving	You could save ₹{x} every month.	Aap har mahine ₹{x} bacha sakte hain.
Close tier	Similar, not identical. Ask your doctor before switching.	Milti-julti hai, bilkul same nahi. Badalne se pehle doctor se poochhein.
Disclaimer	Confirm with your doctor or pharmacist.	Apne doctor ya pharmacist se pakka karein.
Not sure	We could not read this clearly. Please take another photo.	Hum ise saaf nahi padh paaye. Kripya dobara photo lein.

13.4 Accessibility checklist

- Contrast ≥ 4.5:1 for text; do not use colour alone for tier (icon + label too).
- All images and icons have labels; the camera screen is operable with screen readers.
- Text size control (Normal/Large/Extra large); layout never truncates salt names.
- Respect prefers-reduced-motion; no auto-playing audio.

14 · What else to build to make it win

Beyond the baseline, these are ranked by **judge impact ÷ effort**. The recommended set is marked ★. Pick 3–4, not all.

#	Idea	Why it impresses	Effort	Pick
1	**Barcode/QR-first scan** where packs have a code (verify which brands carry one), vision as fallback	Near-100% accuracy for covered brands; shows engineering maturity.	Low–Med	★ if verified
2	**Hindi UI + voice read-aloud**	Directly serves the persona; visible in the demo.	Low	★
3	**Doctor/pharmacist handoff PDF + WhatsApp share**	Embeds the mandatory "confirm" step into a real workflow.	Low	★
4	**Duplicate-salt alert** (two brands, same salt in the basket)	Genuine safety value, unusual in hackathon apps.	Low–Med	★
5	**Public evaluation report** (accuracy, false-positive rate) on S11 and a slide	Credibility; echoes the BODH benchmarking idea.	Low	★
6	WhatsApp entry via n8n	Meets users where they are; uses an Aayu partner.	Med	Stretch
7	Kendra stock crowd-reports ("seen in stock today")	Solves a real pain: trip wasted if out of stock.	Med	Stretch
8	Refill reminders and monthly planner	Retention story.	Low	Stretch
9	ABHA / FHIR MedicationStatement export (sandbox/mock)	Aligns with national digital-health rails and ABDM ecosystem.	Med	Roadmap slide
10	Offline-first scanning (on-device OCR/ONNX)	Works in villages; ASHA persona.	High	Roadmap
11	Handwritten prescription reading	Big wow but high clinical risk; park for after the event.	High	Roadmap
12	Impact dashboard (anonymous): rupees surfaced across scans	Turns a demo into a movement narrative.	Low	Optional

14.1 "Real stuff" checklist for judges

- Public HTTPS URL + QR code on the last slide; judges scan with their own phones.
- Live demo with **actual strips** brought from home (diabetes, BP, cholesterol, thyroid, gastric, pain).
- Show a **real Jan Aushadhi receipt/price photo** next to the app result.
- Show the data page with source names, dates and row counts.
- Show one **honest failure** (an unreadable strip) and how the app handles it. Judges trust teams that show limits.
- Offer the numbers: "₹2,800 → under ₹600; at least ₹26,400 a year for one family."

14.2 Business and scale story (one slide)

- **Users:** chronic-care patients, caregivers, ASHA workers.
- **Distribution:** WhatsApp sharing, Jan Aushadhi Kendras (QR posters), medical college OPDs, NGOs.
- **Sustainability:** free for patients; partnerships with PMBI/Kendras, state health missions, CSR and insurers interested in medicine-cost reduction.
- **Defensibility:** verified brand→salt graph, evaluation data, Hindi voice UX, Kendra stock signals.
- **Roadmap:** more languages, offline mode, ABDM integration, hospital pharmacy integration.

15 · Build plan: roles and the 8-hour schedule

15.1 Roles (adapt to 2–4 people)

Role	Owns	If team is 3	If team is 2
A · Frontend/PWA lead	S01–S12 UI, camera, i18n, PWA, deploy web	Keeps role	Also does design
B · Backend and data lead	FastAPI, DB schema, ETL, matcher, API, deploy API	Keeps role	Also owns eval
C · ML and eval lead	Extractor prompt/schema, resolver, confidence gate, golden set, eval report	Shares with B	Merged into B
D · Product, design, pitch	Figma-lite screens, microcopy, test script, deck, demo, clinician contact	Shared by A and C	Shared

15.2 Pre-event preparation (confirm what is allowed)

- **Safe regardless of rules:** this PRD, wireframes and design tokens, collecting 50–100 strip photos and labels, downloading the PMBI and Kendra lists, curating the top-300 brand→salt sheet, creating accounts and API keys (hosting, DB, vision API, Sentry, maps), rehearsing the pitch, and booking a clinician contact for a 30-minute review.
- **Ask first:** any application code, boilerplate repos, or pre-built models. Aayu's rules say original work built during the event.

15.3 Hour-by-hour

Time	Goal	Deliverables / exit criteria
0:00–0:30	Kickoff and skeleton	Repo, CI, empty web and API deployed to public URLs; env vars set; /health green on phone.
0:30–2:00	Data and extraction	PMBI + brands loaded in Postgres; /v1/scan returns structured JSON on 10 sample strips; camera screen working on a real phone.
2:00–3:30	Resolver and matcher	Brand→salt resolution; EXACT/CLOSE/NONE tiers; savings math; unit tests for edge cases; confirm screen.
3:30–5:00	Result, basket, Kendras	S05 result, S07 list with totals, S08 Kendra finder, disclaimer everywhere; Hindi strings for core flow.
5:00–6:00	Hardening on real devices	Run eval on golden set; fix top 2 error types; quality gate; offline/PWA install; Sentry; keep-warm ping. **Feature freeze at 6:00.**
6:00–7:00	Share, polish, safety review	Share PDF/WhatsApp, voice read-aloud (if time), NTI banner check, clinician spot-check, accessibility pass, Lighthouse.
7:00–8:00	Demo readiness	3× full rehearsal, backup video, QR code, deck final, fallback "demo mode" with cached results, URL stability check.

15.4 Working agreements

- Trunk-based development, small PRs, deploy on every merge to main; one person owns "demo branch" in the last 90 minutes.
- Definition of done: works on a real phone over mobile data, has Hindi string, has loading/error/empty states, has a test or manual check logged.
- Daily-driver rule: all four of you use the deployed URL on your own phones from hour 1.
- Decision log in docs/decisions.md to avoid re-arguing.

16 · Testing and QA

Layer	What to test	How
Unit	Parser (salts, units, release), matcher tiers, savings math, per-unit normalisation	pytest with table-driven cases from section 6.4
Integration	scan → resolve → match end-to-end on 20 strips; error codes	pytest + recorded fixtures; one live test with the vision API
Model eval	Golden set accuracy, false positives, latency	data/eval/eval.py; fail CI if false positives > 0
UI	All 12 screens, loading/error/empty, Hindi, large text	Manual checklist + Playwright smoke tests
Device	3 phones (low/mid/high), Chrome Android, Safari iOS; slow 4G; camera permission flows	Real devices, throttled network
Safety	NTI list behaviour, CLOSE exclusion from totals, disclaimer presence on every screen/PDF	Checklist signed by a clinician contact
Usability	5 non-technical testers (ideally one senior citizen)	Observe, don't help; measure task success and time
Load	10 concurrent scans (demo day with judges)	Simple k6/locust run; set rate limits and queue

16.1 Demo-day failure plan

- Vision API down → OCR/barcode path → manual search. Keep a pre-cached demo mode with 5 scans for guaranteed results.
- Venue Wi-Fi bad → phone hotspot; app shell cached by PWA.
- Backend cold → keep-warm ping every 5 minutes; test 15 minutes before pitch.
- Everything fails → pre-recorded 60-second screen recording and a printed one-page flow.

17 · Running on a phone: deployment checklist

The brief asks for a deployed project that runs on mobile. A PWA gives that without app-store delays.

Step	Detail
Hosting	Web on Vercel/Cloudflare Pages; API on Cloud Run/Render; DB on Neon/Supabase. All HTTPS (camera API requires a secure context).
PWA manifest	name, short_name "Sasti Dawai", icons 192/512 (maskable), theme_color #063257, display standalone, start_url /.
Service worker	Precache app shell + i18n + kendra cache; runtime cache for GETs; show an offline page; queue failed scans.
Camera on iOS/Android	Test getUserMedia on Safari iOS and Chrome Android; provide *input type=file capture=environment* fallback.
Performance	Code-split screens; lazy-load map; compress images client-side; font subsetting; Lighthouse mobile ≥ 85.
Domain and QR	Short memorable URL (or a free subdomain); generate a QR for the deck and a printed A5 card for the table.
Backup APK (optional)	Wrap with Capacitor to give an installable APK if time permits.
Config	Environment variables per stage; secrets never in the repo; separate demo and dev databases.
Pre-demo smoke test	Scan 5 real strips on 3 phones over mobile data; check /health; check price_date is today.

18 · Demo script and pitch

18.1 Three-minute live demo

Time	Say	Show
0:00–0:25	"Mohanlal spends ₹2,800 a month on diabetes and BP medicines. The same salts at a Jan Aushadhi Kendra cost under ₹600. He never knew."	Slide: the scene, big numbers
0:25–1:30	"Watch. I scan his strip." (read steps aloud) "It reads the brand, finds the salt, verifies it, and checks the exact equivalent."	Live phone scan → Confirm → Result (green Exact tier, ₹ saving)
1:30–2:00	"He builds his monthly list. ₹ saved per month, ₹ per year." "It warns if two brands hide the same salt."	My monthly list + duplicate alert
2:00–2:25	"It never decides for him: close matches are labelled, critical drugs say do not switch. Every screen says confirm with your doctor or pharmacist."	NTI example + disclaimer + share-with-doctor PDF
2:25–2:45	"Here's our accuracy on 80 real strips, and one failure we handle honestly."	S11 eval + unreadable strip example
2:45–3:00	"Scan this QR and try it on your own phone."	QR + URL on screen

18.2 Idea-deck outline (Round 1; use Aayu's .pptx template)

- 1 · Title and one-line promise
- 2 · The scene: Mohanlal and the numbers
- 3 · Why it happens (brand vs salt, 2-minute consults)
- 4 · Solution overview and user flow
- 5 · Exact-match safety model (tiers)
- 6 · Architecture and ML pipeline
- 7 · Data: Jan Aushadhi catalogue, provenance, freshness
- 8 · Differentiation vs Jan Aushadhi Sugam and others
- 9 · Impact, scale and roadmap (ABDM)
- 10 · Team and 8-hour plan

18.3 Likely judge questions and crisp answers

Question	Answer
How is this different from the Jan Aushadhi Sugam app?	It starts from the strip, not from a typed salt. Scan → verified salt → exact-match tiers → monthly basket → shareable doctor summary, in Hindi with voice.
Same salt means same effect?	We only claim equivalence for same salt set, strength, form and release; critical and special forms are flagged; and every result says to confirm with a doctor or pharmacist.
How accurate is the scan?	Show eval numbers; explain confidence gating and the confirm step; zero false positives on our set.
Why not just ask an LLM?	LLMs can hallucinate. We use ML for reading, but equivalence and prices come from verified data and rules.
Is the generic really good quality?	We cite the government's stated process (WHO-GMP sourcing, NABL batch testing) and avoid over-claiming.
What about privacy?	No accounts, no patient data, photos discarded by default, minimal processors listed.
Can it scale?	Stateless API, managed Postgres, nightly ETL; one dataset covers the whole country; distribution through Kendras and WhatsApp.
Business model?	Free for patients; partnerships with PMBI, state health missions, NGOs, insurers; optional Kendra analytics.

19 · Risks and mitigations

Risk	Likelihood / impact	Mitigation
Rules disallow pre-written code	Med / High	Ask organisers now; limit pre-event prep to data, design, accounts; plan to write code on-site.
Vision model misreads strength or release type	Med / High	Schema + cross-check against DB; confidence gate; confirm screen; eval-driven prompt fixes; barcode first.
Brand→salt mapping errors	Med / High	Curate top-300 by hand; clinician review; mismatch blocks matching.
PMBI list format or availability changes	Low / Med	Download early; store raw copy with checksum; defensive parser.
No Wi-Fi or API outage at venue	Med / High	Hotspot, cached demo mode, OCR fallback, recorded video.
Camera permissions or iOS quirks	Med / Med	Test early on real devices; file-input fallback.
Unsafe suggestion presented	Low / Critical	Exact-only savings; NTI list; post-filter on explainer; clinician check; one-tap reporting.
Scope creep	High / Med	Cut line at hour 5; feature freeze at hour 6; MoSCoW enforced by product owner.
Cold starts / free-tier limits	Med / Med	Keep-warm pings; paid tier for demo day if needed; load test.
Legal/regulatory perception (looks like medical advice)	Low / High	Clear positioning as an information tool; disclaimer everywhere; no dosing; no purchase flow; mentor review.

20 · Open decisions (resolve in the first 24 hours)

#	Decision	Owner	Default if undecided
1	What pre-event work do Aayu rules allow?	Product lead (email organisers)	Prep data/design only; code on-site
2	Vision/LLM provider and key budget	ML lead	Gemini Flash-class model with JSON schema
3	Which QR/barcode coverage can we verify on real packs?	ML lead	Treat as stretch; vision-first
4	NTI/critical drug list and clinician reviewer	Product lead	Conservative list from standard references, reviewed by contact doctor
5	Hosting accounts and region (India-near for latency)	Backend lead	Cloud Run Mumbai + Neon/Supabase
6	Which 5 strips form the demo set (diabetes, BP, lipid, thyroid, gastric)?	Everyone	Choose by availability of exact Jan Aushadhi match
7	Do we show other low-cost brands or only Jan Aushadhi?	Product lead	Jan Aushadhi only
8	Team size and role split	Everyone	See section 15.1

Appendix A · Matching pseudocode

```
def match(scanned: CanonicalProduct, units_per_month: int) -> MatchResult:
    if scanned.confidence < CONF_THRESHOLD or scanned.crosscheck_failed:
        return needs_confirmation()
    exact, close = [], []
    for g in generics.where(route=scanned.route, dosage_form=scanned.dosage_form):
        if salt_set(g) == salt_set(scanned) and strengths_equal(g, scanned):
            if g.release_type == scanned.release_type:
                exact.append(g)              # all five dimensions equal
            else:
                close.append((g, "release type differs"))
        elif salt_moiety_set(g) == salt_moiety_set(scanned):
            close.append((g, "strength or salt form differs"))
    warnings = []
    if any(s.is_nti for s in scanned.salts):
        warnings.append("NTI: do not switch without your doctor")
    results = sorted(exact, key=lambda g: g.mrp / g.pack_count)
    saving = [(unit_price(scanned) - unit_price(g)) * units_per_month for g in results]
    tier = "exact" if results else ("close" if close else "none")
    return MatchResult(tier, results, close, saving if tier == "exact" and not warnings_block_savings(warnings) else None,
                       disclaimer=FIXED_DISCLAIMER, price_date=data_version.price_date)
```

Appendix B · Seed list of therapeutic areas to curate first

Focus the first ~300 brands on chronic, high-spend, high-volume conditions (matches the Mohanlal persona and gives the most demo value). Have a clinician contact confirm the final list and the NTI flags.

Area	Typical salts to cover (examples)	Notes
Diabetes	Metformin (IR and SR), glimepiride, gliclazide, vildagliptin, sitagliptin, teneligliptin, pioglitazone, and common combinations	Watch SR vs IR and fixed-dose combinations
Hypertension and heart	Amlodipine, telmisartan, losartan, olmesartan, atenolol, metoprolol, ramipril, chlorthalidone, and combinations	Combinations are common; check strengths carefully
Lipids	Atorvastatin, rosuvastatin, fenofibrate, ezetimibe, and combinations	Strength variants (5/10/20/40 mg)
Thyroid	Levothyroxine, thyroxine	**NTI: do not switch without doctor**
Blood thinners and antiplatelets	Aspirin, clopidogrel, warfarin	Warfarin: NTI/critical
Gastric	Pantoprazole, rabeprazole, omeprazole, domperidone, and combinations	Delayed-release and enteric-coated forms
Pain and inflammation	Paracetamol, diclofenac, aceclofenac, ibuprofen, and combinations	Many combinations
Antibiotics and basics	Amoxicillin (+ clavulanic acid), azithromycin, cefixime, metronidazole, cetirizine, montelukast	Short courses: no monthly saving
Neuro / psychiatric (limited)	Phenytoin, carbamazepine, levetiracetam, escitalopram	Anti-epileptics: NTI/critical; treat conservatively

Appendix C · References

- Aayu 2026 Problem statements: aayu.devaarambh.com/problems
- Aayu 2026 Rules: aayu.devaarambh.com/rules
- Aayu 2026 About (problem framing, ABHA/ABDM/SAHI/BODH): aayu.devaarambh.com/about
- Drishti IAS, "Uttar Pradesh Leads India with 4,226 Operational Jan Aushadhi Kendras" (1 Aug 2026): basket of 2,110 medicines and 315 surgical lines; WHO-GMP sourcing, NABL testing
- Drishti IAS, "25,000 Jan Aushadhi Kendras by March 2027" (17 Dec 2025)
- Kashmir Life, report on Lok Sabha reply: 17,990 Kendras as of 31 Dec 2025
- Swarajya, "Rs 30,000 Crore Saved Via Jan Aushadhi Kendras" (28 Mar 2025): 50–80% lower, fixed MRP, supply network
- IAS Gyan, Jan Aushadhi Sugam app summary (BPPI / Department of Pharmaceuticals)
- PMBI official site and product list (pmbi.co.in, janaushadhi.gov.in): primary data source; re-verify before launch

Verify before presenting: counts and prices change. Re-pull the PMBI list, confirm Kendra totals on the official site, and have a clinician review the NTI list and the demo medicines. This PRD is a product plan, not medical or legal advice.